
SENTIMENT ANALYSIS USING NLP

A
PROJECT REPORT

Submitted to

**Laxman Devram Sonawane College of Arts and Commerce,
Kalyan(west), 421301**

In partial fulfillment for the award of the degree of

**Master of Science
In
Information Technology**



(Affiliated to University of Mumbai)

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UNDER THE GUIDENCE OF

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INFORMATION
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**Laxman Devram Sonawane
College of Arts, Commerce and
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The Kalyan Wholesale Merchants' Education Society's,
LAXMAN DEVRAM SONAWANE DEGREE COLLEGE
Department of Information Technology

M.Sc. Part - II

Certificate

This is to certify that the project entitled “**SENTIMENT ANALYSIS** ” is a Bonafide work of “**Rhishi J. Nair**”, **Seat No: 1314229** submitted to the Laxman Devram Sonwane College Affiliated to University of Mumbai in partial fulfilment of the requirement for the award of the degree of “**Postgraduate**” in “**Master of Science in Information Technology**” during the academic year 2025- 2026.

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DECLARATION

I, **Rhishi J. Nair** the student of Master's of Science in Information Technology, declare that the work entitled "**Sentiment Analysis**" has been successfully completed under the guidance of **DR. NRUPURA DIXIT**. This dissertation work is submitted in partial fulfilment of the requirements for the award of Degree of Master of Science in Information Technology during the academic year 2025- 2026. Further the matter embodied in the project report has not been submitted previously by anybody for the award of any degree or diploma to any university.

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Title of Research Project : Sentiment Analysis using NLP

Brief description of Project work carried out : Focuses on identifying and classifying emotions, attitudes, and opinions expressed in textual data.

Year of completion of Research Project : 2025-26

Was your project work experience related to your major area of study?

- Yes, to a large degree
- Yes, to a slight degree
- No, not related at all

Indicate the degree to which you agree or disagree with the following statements

This experience has:	Strongly Agree	Agree	No opinion	Disagree	Strongly Disagree
Given me the opportunity to explore a career field					
Allowed me to apply classroom theory to practice					
Helped me develop my decision-making and problem-solving skills					
Expanded my knowledge about the work world before permanent employment					
Helped me develop my written and oral communication skills					
Provided a chance to use leadership skills (influence others, develop ideas with others, stimulate decision-making and action)					
Expanded my sensitivity to the ethical implications of the work involved					
Made it possible for me to be more					

confident in new situations					
Given me a chance to improve my interpersonal skills					
Helped me learn to handle responsibility and use my time wisely					
Helped me discover new aspects of myself that I didn't know existed before					
Helped me develop new interests and abilities					
Helped me clarify my career goals					
Allowed me to acquire information and/ or use equipment not available at my Institute					
Allowed me to realize socioeconomic/environmental issues.					

ABSTRACT

Sentiment Analysis, also referred to as opinion mining, is a key application of Natural Language Processing (NLP) that focuses on identifying and classifying emotions, attitudes, and opinions expressed in textual data. With the widespread use of social media, online platforms, customer feedback systems, and digital communication, vast amounts of unstructured text are generated every day. Analysing this data manually is impractical, making automated sentiment analysis an essential tool for extracting meaningful insights.

Sentiment analysis techniques aim to determine whether a given text conveys a positive, negative, or neutral sentiment by applying NLP and machine learning methods. The process typically involves text pre-processing, feature extraction, and sentiment classification using statistical or learning-based models. These techniques help organizations understand public opinion, customer satisfaction, brand perception, and market trends in an efficient and scalable manner.

Overall, sentiment analysis plays a vital role in decision-making across various domains such as business intelligence, marketing, politics, healthcare, and social research. Its ability to transform raw textual data into actionable insights highlights the growing importance of NLP-driven analytics in the modern data-driven world.

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CHAPTER 1

1.1 INTRODUCTION

Sentiment Analysis, also referred to as opinion mining, is an important application of Natural Language Processing (NLP) that focuses on identifying and extracting subjective information from textual data. It is used to determine whether a piece of text expresses a positive, negative, or neutral sentiment. In today's digital world, people constantly share their opinions through online platforms such as e-commerce websites, social media, blogs, and feedback systems. These opinions play a crucial role in shaping business strategies, improving products, and understanding customer satisfaction.

With the rapid expansion of online shopping platforms like Amazon, millions of product reviews are generated daily. These reviews contain valuable insights about customer experiences, preferences, and expectations. However, manually analysing such a large volume of data is not practical. Therefore, automated sentiment analysis systems are required to process and analyse textual data efficiently and accurately. By using machine learning and deep learning techniques, sentiment analysis helps organizations make data-driven decisions and enhance user experience.

This project aims to develop a Sentiment Analysis system that classifies Amazon product reviews into positive and negative categories. The dataset used in this project consists of product review text and corresponding rating scores. To improve the quality of the data, several preprocessing steps are performed, including removing neutral reviews, handling missing values, text normalization, and converting text into a suitable numerical format. Tokenization and padding techniques are applied to ensure that the text data can be effectively processed by the neural network model.

For sentiment classification, a Long Short-Term Memory (LSTM) based deep learning model is implemented. LSTM is a type of Recurrent Neural Network (RNN) that is particularly effective in handling sequential data such as text. It is capable of capturing contextual information and long-term dependencies between words, which makes it suitable for understanding the sentiment expressed in reviews. The model is trained using labeled data and optimized using appropriate loss functions and evaluation metrics.

The performance of the model is evaluated using accuracy, training and validation loss, confusion matrix, and classification report. These metrics help in understanding how well the model predicts sentiments on unseen data. The final system is capable of taking a new product review as input and predicting whether the sentiment expressed is positive or negative. This project demonstrates how NLP and deep learning techniques can be effectively combined to analyse large-scale textual data and extract meaningful insights from customer opinions.

1.2 OBJECTIVES

1. **To design and develop a sentiment analysis system** that automatically classifies Amazon product reviews into positive and negative sentiments using Natural Language Processing techniques.
2. **To collect and preprocess real-world Amazon review data** by removing neutral reviews, handling missing values, and cleaning textual content to improve model accuracy.
3. **To convert textual review data into numerical representations** using tokenization and padding techniques suitable for deep learning models.
4. **To implement an LSTM-based deep learning model** for capturing contextual and sequential information present in customer reviews.

5. **To train and evaluate the model** using appropriate performance metrics such as accuracy, loss, confusion matrix, and classification report.
6. **To analyse the effectiveness of the proposed model** in predicting sentiment on unseen test data.
7. **To build a prediction function** that takes new user input and accurately determines whether the sentiment expressed is positive or negative.
8. **To demonstrate the practical application of NLP and deep learning** in extracting meaningful insights from large-scale e-commerce review data.

1.3 PROBLEM STATEMENT

E-commerce platforms like Amazon generate a large number of customer reviews that contain valuable opinions about products and services. Analyzing these reviews manually is inefficient and impractical due to the massive volume and unstructured nature of textual data. Traditional methods often fail to accurately capture the contextual meaning of customer feedback.

This project addresses the need for an automated sentiment analysis system that can process and classify Amazon product reviews into positive or negative sentiments. By using Natural Language Processing techniques and an LSTM-based deep learning model, the system aims to accurately understand customer opinions and provide meaningful insights from large-scale review data.

1.4 METHODOLOGY

The methodology of this project follows a systematic approach to build an effective

sentiment analysis system using Natural Language Processing and deep learning techniques. The complete process is divided into the following steps:

1. Data Collection

A real-world Amazon product reviews dataset is used for this project. The dataset contains customer review text along with rating scores. These reviews serve as the primary input for sentiment classification.

2. Data Preprocessing

To improve data quality and model performance, several preprocessing steps are applied:

- Removal of neutral reviews (rating score = 3)
- Handling missing and null values
- Text cleaning such as lowercasing and removal of unwanted characters

After preprocessing, sentiment labels are assigned where ratings of 4 and 5 are considered positive and ratings of 1 and 2 are considered negative.

3. Text Tokenization and Padding

The cleaned review text is converted into numerical form using tokenization. Each word is mapped to a unique integer. Since reviews have different lengths, padding is applied to ensure all sequences have equal length for model input.

4. Train–Test Split

The dataset is divided into training and testing sets using an 80:20 ratio. The training set is used to train the model, while the testing set is used to evaluate its performance on unseen data.

5. Model Building

An LSTM-based deep learning model is developed using an embedding layer, LSTM layer, dropout layer, and dense output layer with a sigmoid activation function. This architecture helps capture contextual and sequential information present in textual data.

6. Model Training

The model is trained using the training dataset with binary cross-entropy as the loss function and Adam optimizer. Training and validation accuracy and loss are monitored over multiple epochs to avoid overfitting.

7. Model Evaluation

The trained model is evaluated using accuracy, confusion matrix, and classification report. These metrics help assess the effectiveness and reliability of the sentiment classification system.

8. Sentiment Prediction

A prediction function is implemented to allow the system to classify new, unseen product reviews as either positive or negative, demonstrating the practical application of the model.

1.5 SCOPE

The scope of this project is to design and implement an automated sentiment analysis system that classifies Amazon product reviews into positive and negative sentiments using Natural Language Processing and LSTM-based deep learning techniques. The system enables efficient analysis of large-scale customer feedback and supports better understanding of user opinions. The proposed model can be extended to handle neutral sentiments, larger datasets, real-time review analysis, and reviews from other domains such as social media and online services.

CHAPTER 2

2.1 HISTORY

The evolution of Natural Language Processing (NLP) and Sentiment Analysis has been gradual and spans several decades. Initially, computers were designed only for numerical calculations and had no capability to understand or interpret human language. The idea of enabling machines to process natural language emerged as researchers attempted to bridge the gap between human communication and machine understanding.

Early Developments (1950s – 1960s)

The foundations of NLP were laid during the 1950s, when researchers began experimenting with language translation and basic text processing. Early systems were rule-based, meaning they relied on manually written grammar rules and dictionaries. These systems worked only in controlled environments and failed to handle ambiguity, context, or variation in human language.

During the 1960s, linguistic theories such as syntax and semantics were introduced into computational models. Although these systems could process structured sentences, they lacked the ability to understand meaning, emotions, or opinions expressed in text. At this stage, sentiment analysis did not exist as a separate research area, but the linguistic groundwork was established.

Rule-Based and Knowledge-Based Era (1970s)

In the 1970s, NLP research received significant support from organizations such as DARPA. Most language processing systems during this period were still rule-based and depended

heavily on expert-defined knowledge. These systems required extensive manual effort and were difficult to scale.

Although effective for small datasets, rule-based approaches were not suitable for real-world text, where language is informal, ambiguous, and context-dependent. This limitation highlighted the need for more flexible and automated approaches, paving the way for statistical methods.

Statistical NLP Era (1980s)

The 1980s marked a major shift from rule-based systems to statistical and probabilistic approaches. Researchers began using mathematical models to analyze language patterns based on probability rather than fixed rules. One of the most important breakthroughs of this decade was the introduction of the Hidden Markov Model (HMM).

HMM allowed systems to estimate the probability of word sequences and handle uncertainty in language. This approach significantly improved performance in text classification tasks and laid the foundation for automated sentiment detection. Vocabulary handling and pattern recognition became more robust during this era.

Machine Learning Era (1990s)

In the 1990s, the availability of faster computers and large digital text collections accelerated NLP research. Machine learning algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Decision Trees were widely applied to text classification problems.

During this period, sentiment analysis emerged as a distinct research field. Researchers started analyzing movie reviews, product reviews, and customer feedback to classify opinions as positive or negative. Techniques such as Bag of Words (BoW) were used to

convert text into numerical form. However, these models ignored word order and context, limiting their understanding of sentiment in complex sentences.

Web and E-Commerce Expansion (2000s)

The 2000s witnessed a massive growth of the internet, social media, and e-commerce platforms like Amazon. This led to the availability of large-scale user-generated content such as reviews, ratings, and comments. Sentiment analysis became highly relevant for business intelligence and decision-making.

During this decade, feature-based methods like TF-IDF, n-grams, and lexicon-based approaches gained popularity. While these methods improved accuracy, they still relied on manual feature engineering and struggled with sarcasm, negation, and contextual meaning.

Deep Learning Revolution (2010s)

A major breakthrough occurred in the 2010s with the rise of deep learning. Neural network-based models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks transformed sentiment analysis. Unlike traditional models, LSTMs could process text sequentially and capture long-term dependencies between words.

LSTM models significantly improved sentiment classification accuracy, especially for long and complex reviews such as those found on Amazon. The availability of large datasets, GPUs, and deep learning frameworks enabled efficient training of such models. Sentiment analysis systems became more accurate, scalable, and reliable during this period.

Modern NLP and Current Trends (2020s)

In recent years, transformer-based models and pre-trained language models have further advanced NLP capabilities. However, LSTM-based sentiment analysis models remain

widely used in academic projects and real-world applications due to their balance between performance and computational efficiency.

Today, sentiment analysis plays a crucial role in customer feedback analysis, product recommendation systems, brand monitoring, and decision support systems. The continuous evolution of NLP techniques has made it possible to extract meaningful insights from large volumes of unstructured textual data

THE FUTURE

The future of sentiment analysis is highly promising due to the rapid growth of digital communication and continuous advancements in Natural Language Processing (NLP) and Artificial Intelligence (AI). As individuals and organizations increasingly rely on online platforms to express opinions, emotions, and feedback, sentiment analysis will play a crucial role in understanding human behaviour and decision-making processes.

One of the most significant future developments in sentiment analysis will be the improvement in contextual understanding. Traditional models mainly focus on identifying positive or negative words, whereas future systems will be able to understand context, intent, sarcasm, and irony more accurately. Advanced deep learning models will analyze not only individual words but also sentence structure, tone, and surrounding context to determine the true sentiment expressed by users.

Another important direction is the expansion of multilingual and cross-lingual sentiment analysis. In the future, sentiment analysis systems will be capable of processing and understanding text from multiple languages simultaneously. This will help global organizations analyse customer feedback from different regions without language barriers. Improved translation models and language-independent embeddings will enhance sentiment detection across diverse linguistic communities.

The future will also see increased integration of real-time sentiment analysis. With the growth of social media, live reviews, and instant feedback platforms, businesses will be able to monitor public sentiment in real time. This will enable faster response to customer concerns, brand reputation management, and proactive decision-making. Real-time sentiment analysis will become especially important in areas such as stock market analysis, political campaigns, and crisis management.

Another major advancement will be the use of emotion and aspect-based sentiment analysis. Instead of classifying text into only positive or negative categories, future systems will identify multiple emotions such as happiness, anger, sadness, fear, and trust. Additionally, aspect-based sentiment analysis will allow systems to determine sentiment towards specific product features, such as price, quality, or performance, providing more detailed and actionable insights.

With the rapid growth of deep learning and neural networks, future sentiment analysis models will become more accurate and efficient. Architectures such as transformer-based models, hybrid deep learning systems, and attention mechanisms will further enhance sentiment prediction. However, models like LSTM will continue to remain relevant due to their efficiency in handling sequential data and their suitability for academic and industrial applications.

The integration of sentiment analysis with other technologies will also shape its future. Combining sentiment analysis with speech recognition, image analysis, and video processing will enable multi-modal sentiment analysis. This will help systems understand emotions not only from text but also from voice tone, facial expressions, and visual cues, leading to more human-like understanding.

Ethical considerations and data privacy will become increasingly important in the future. Researchers and developers will focus on building fair, unbiased, and transparent sentiment

analysis systems. Responsible AI practices will ensure that sentiment analysis models do not misinterpret user opinions or reinforce social bias.

In conclusion, the future of sentiment analysis lies in building intelligent, context-aware, multilingual, and real-time systems that can understand human emotions more accurately. As digital data continues to grow, sentiment analysis will remain a key tool for businesses, governments, and researchers to extract meaningful insights and make informed decisions.

2.2 ANALYSIS

Sentiment Analysis has become an essential part of modern data analytics due to the rapid growth of digital platforms such as e-commerce websites, social media, blogs, and online feedback systems. From customer reviews on Amazon to opinions shared on social media, users continuously express their emotions and experiences in textual form. Analysing this large volume of unstructured text manually is inefficient and impractical, which creates a strong need for automated sentiment analysis systems.

This Sentiment Analysis project focuses on analysing customer opinions using a real-world Amazon product reviews dataset. The project utilizes Natural Language Processing (NLP) techniques along with a deep learning model based on Long Short-Term Memory (LSTM) to classify reviews into positive and negative sentiments. The implementation is carried out using Python, Keras, and TensorFlow, enabling efficient model training and accurate sentiment prediction on unseen textual data.

2.2.1 Data Ingestion and Processing

Data ingestion and preprocessing are the most crucial steps in sentiment analysis, similar to preprocessing in image or speech recognition systems. Since machine learning models

cannot directly understand raw text, the textual data must be cleaned, transformed, and converted into numerical format.

The Amazon reviews dataset is first loaded and filtered to remove unnecessary or noisy data. Neutral reviews are eliminated to improve classification clarity. Text preprocessing steps include removing missing values, converting text to lowercase, and eliminating unwanted characters. These steps help in improving model performance and reducing computational complexity.

```
import pandas as pd
import numpy as np
import nltk
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense, Dropout

nltk.download('stopwords')

[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
True
```

Text Representation and Tokenization

In sentiment analysis, textual data is transformed into numerical form using tokenization, where each word is mapped to a unique integer. Since reviews vary in length, padding is applied to ensure uniform input size for the deep learning model.

This transformation allows the LSTM model to process text sequentially and learn relationships between words, enabling it to capture contextual and semantic information present in customer reviews.

2.3 Types of Sentiment Analysis

Sentiment Analysis systems can be classified into different types based on their functionality and level of detail.

1. Polarity-Based Sentiment Analysis

This is the most basic form of sentiment analysis, where text is classified into:

- Positive
- Negative
- (Sometimes Neutral)

This project focuses on binary polarity classification, identifying whether a review expresses positive or negative sentiment.

2. Subjectivity-Based Sentiment Analysis

This type distinguishes between:

- Subjective text (opinions, emotions)
- Objective text (facts, information)

Subjective text is more relevant for sentiment analysis as it reflects user opinions.

3. Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis identifies sentiment towards specific features or aspects of a product, such as price, quality, or performance. Although not implemented in this project, it represents a possible extension.

4. Emotion-Based Sentiment Analysis

This approach goes beyond polarity and identifies emotions such as happiness, anger, sadness, or frustration. Advanced deep learning models are commonly used for this purpose.

2.4 Sentiment Analysis Process

The sentiment analysis process followed in this project includes the following steps:

1. Data Collection – Collecting Amazon product reviews and rating scores.
2. Data Preprocessing – Cleaning text data and removing neutral reviews.
3. Tokenization and Padding – Converting text into numerical sequences.
4. Model Training – Training an LSTM-based deep learning model.
5. Model Evaluation – Evaluating performance using accuracy, loss, confusion matrix, and classification report.
6. Sentiment Prediction – Predicting sentiment for new user-input reviews.

2.5 Applications of Sentiment Analysis

Sentiment analysis has wide applications across multiple domains:

I. Commercial Perspective

Used in e-commerce platforms to analyse customer reviews, improve products, and enhance user experience.

II. Social Media Analysis

Helps in understanding public opinion, brand reputation, and trending topics.

III. Educational Perspective

Assists in analysing student feedback and improving course content.

IV. Business Intelligence

Supports decision-making by extracting insights from customer opinions and feedback.

CHAPTER 3

3.1 System Design and Implementation

Sentiment Analysis is a key application of Natural Language Processing (NLP) that focuses on identifying and classifying emotions or opinions expressed in textual data. In this project, a sentiment analysis system is developed to analyze Amazon product reviews and classify them into positive or negative sentiments using a deep learning model based on LSTM (Long Short-Term Memory).

With the rapid growth of e-commerce platforms, customer reviews play a critical role in influencing purchasing decisions. However, due to the massive volume of reviews generated daily, manual sentiment evaluation is not feasible. Therefore, this project aims to automate sentiment classification using machine learning and deep learning techniques implemented in Python using TensorFlow and Keras.

3.1.1 Tools and Libraries Used

This section describes the Python libraries and frameworks used for implementing the sentiment analysis system. TensorFlow and Keras are used for building the LSTM model, while Pandas and Scikit-learn are used for data handling and evaluation.

```
python
```

```
!pip install nltk tensorflow pandas matplotlib scikit-learn
```

```
import pandas as pd
```

```
import numpy as np
```

```
import nltk
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
from tensorflow.keras.preprocessing.text import Tokenizer
```

```
from tensorflow.keras.preprocessing.sequence import pad_sequences
```

```
from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Embedding, LSTM, Dense, Dropout
```

```
nltk.download('stopwords')
```

3.1.2 Dataset Loading

The Amazon product reviews dataset is loaded from Google Drive. Only review text and rating score columns are selected for sentiment analysis.

```
python
```

```
from google.colab import drive
```

```
drive.mount('/content/drive')
```

```
df = pd.read_csv('/content/drive/MyDrive/data/amazon_reviews.csv')
```

```
df = df[['Text', 'Score']]
```


3.1.3 Data Processing

Neutral reviews are removed to improve classification clarity. Sentiment labels are created where ratings ≥ 4 are positive and ≤ 2 are negative.

```
python

df = df.sample(60000, random_state=42)

# Remove neutral reviews
df = df[df['Score'] != 3]

# Create sentiment column
df['sentiment'] = df['Score'].apply(lambda x: 1 if x >= 4 else 0)

# Rename column
df.rename(columns={'Text': 'text'}, inplace=True)

# Remove null values
df.dropna(inplace=True)

df.head()
```

3.1.4 Train and Test Split

The dataset is split into training (80%) and testing (20%) sets to evaluate model performance on unseen data.

```
python

X = df['text']
y = df['sentiment']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

3.1.5 Text Tokenization and Padding

Tokenization converts text into numerical sequences. Padding ensures uniform input length for the LSTM model.

```
python

tokenizer = Tokenizer(num_words=10000)
tokenizer.fit_on_texts(X_train)

X_train_seq = tokenizer.texts_to_sequences(X_train)
X_test_seq = tokenizer.texts_to_sequences(X_test)

max_len = 200
X_train_pad = pad_sequences(X_train_seq, maxlen=max_len)
X_test_pad = pad_sequences(X_test_seq, maxlen=max_len)
```

3.1.6 LSTM Model Architecture

An LSTM-based deep learning model is designed to capture sequential and contextual information from text reviews.

```
python

model = Sequential([
    Embedding(10000, 128, input_length=max_len),
    LSTM(128),
    Dropout(0.5),
    Dense(1, activation='sigmoid')
])

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

3.1.7 Model Training

The model is trained using the training dataset with validation to monitor overfitting.

```
python

history = model.fit(
    X_train_pad, y_train,
    epochs=5,
    batch_size=64,
    validation_split=0.1
)
```

3.1.8 Model Evaluation

Model performance is evaluated using accuracy and classification report.

```
python

y_pred = (model.predict(X_test_pad) > 0.5).astype(int)

acc = accuracy_score(y_test, y_pred) * 100
print(f"Test Accuracy: {acc:.2f}%")

print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

3.1.9 Confusion Matrix Visualization

Confusion Matrix for Sentiment Classification

```
python
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
plt.imshow(cm)
plt.title("Confusion Matrix")
plt.colorbar()
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

3.1.10 Training and Validation Graph

Accuracy and Loss Curves

```
python
```

```
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.legend()
plt.show()

plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.legend()
plt.show()
```

3.1.11 Sentiment Prediction on New Reviews

This function allows prediction of sentiment for new user-entered reviews.

```
python
```

```
def predict_sentiment(text):
    seq = tokenizer.texts_to_sequences([text])
    pad = pad_sequences(seq, maxlen=max_len)
    prediction = model.predict(pad)[0][0]
    return "Positive" if prediction > 0.5 else "Negative"

print(predict_sentiment("This product is amazing and tastes great"))
print(predict_sentiment("Worst product I have ever bought"))
```

3.1.12 Dataset Split Summary

python

```
total_samples = len(df)
train_samples = len(X_train)
test_samples = len(X_test)

print("Train-Test Split Information")
print(f"Total Samples : {total_samples}")
print(f"Training Data : {train_samples}")
print(f"Testing Data : {test_samples}")
```

3.2 System Development Approach

3.2.1 Activity Diagram

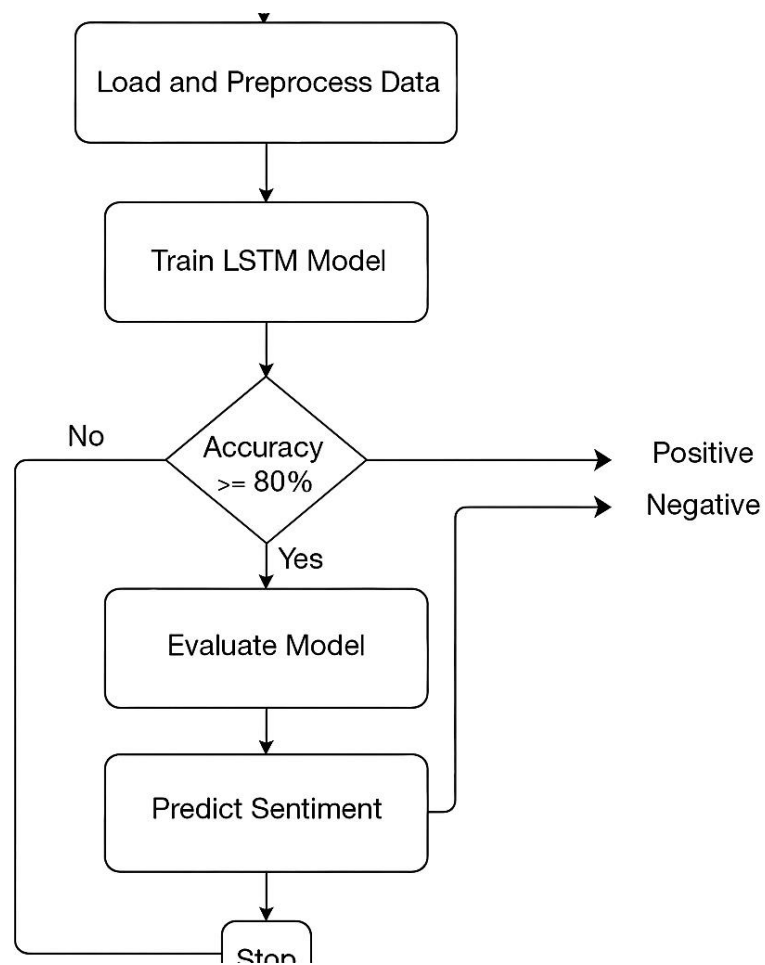
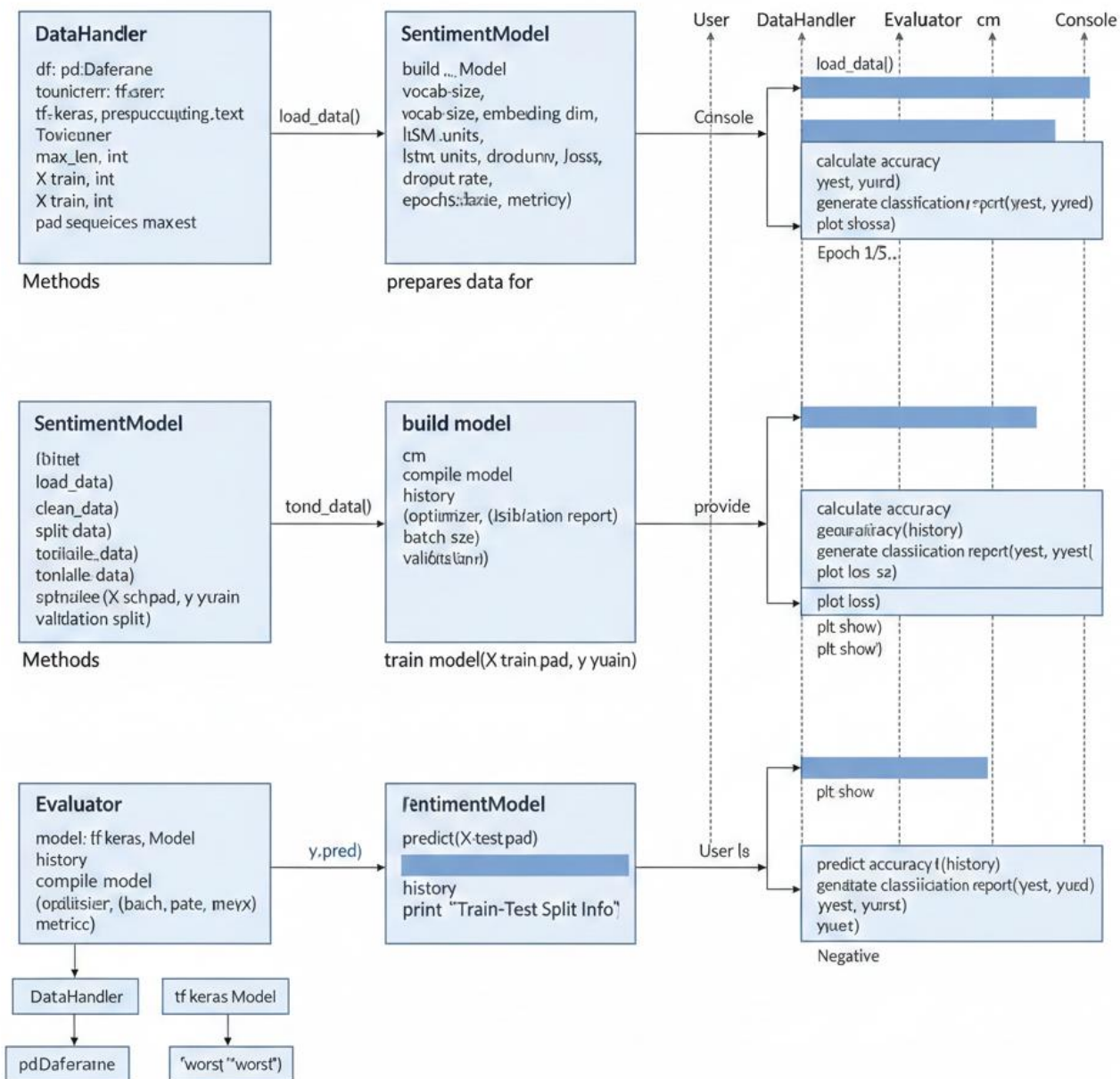


Fig 3.1 Activity Diagram

3.2.2 Class Diagram

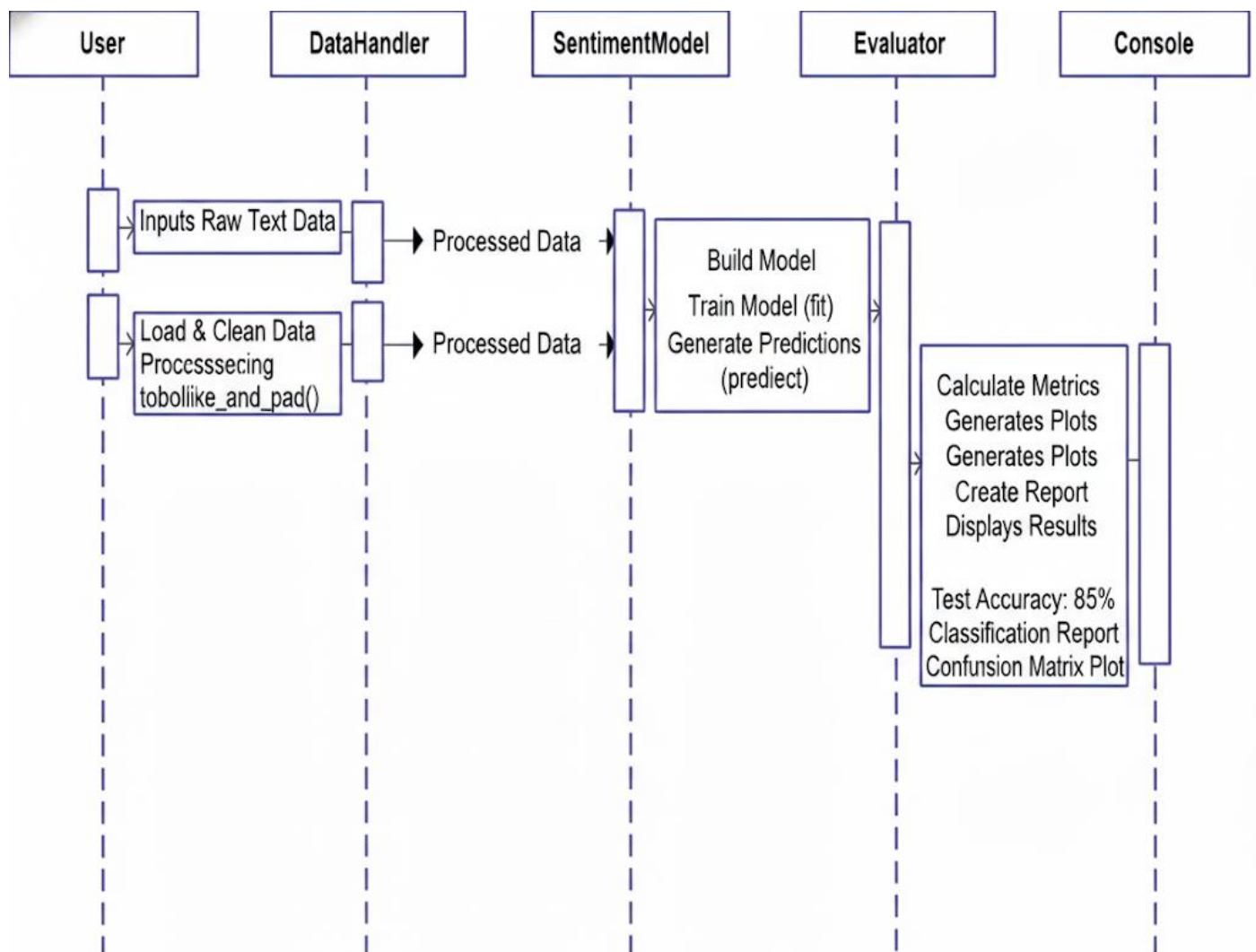
Sentiment Analysis Project - Class Diagram



Tuesday, January 27, 2026 at 1:07:27 AM IST

Fig 3.2 Class Diagram

3.3.3 Sequence Diagram



Tuesday, January 27, 2026 at 1:2:00 AM IST 🌄

Fig 3.3 Sequence Diagram

CHAPTER 4

(PERFORMANCE ANALYSIS)

4.1 System Requirements

The Sentiment Analysis system is implemented using Machine Learning and Natural Language Processing techniques. It requires moderate computational resources for data preprocessing, model training, and prediction. The system is designed to efficiently analyze large volumes of textual data such as online product reviews and classify them into positive or negative sentiments.

4.1.1 Minimum Requirements

- a. Processor with minimum 1.6 GHz speed
- b. 4 GB RAM
- c. Minimum 10 GB free storage
- d. Stable internet connection (for dataset access and library installation)
- e. Any basic keyboard and display device

These minimum requirements are sufficient for running small datasets, basic preprocessing, and testing the sentiment analysis model.

4.1.2 Recommended (Best) Requirements

- a. 2.4 GHz or higher multi-core processor
- b. 8 GB RAM or more
- c. SSD storage for faster data access
- d. Python-supported GPU (optional, for faster training)
- e. Less than 15% memory consumption per process

The recommended configuration improves model training speed, especially while working with deep learning models like LSTM, and large datasets such as Amazon product reviews.

4.2 Hardware Requirements

Processor (CPU)

The performance of the Sentiment Analysis system highly depends on the processor speed. Text preprocessing, tokenization, sequence padding, and LSTM model training require efficient computation. A faster processor significantly reduces training time and improves overall system responsiveness.

Memory (RAM)

RAM plays a crucial role while handling large textual datasets and during neural network training. Insufficient memory can slow down execution and cause system crashes. Therefore, higher RAM ensures smooth processing of data batches and model evaluation.

Storage

Adequate storage is required to store datasets, trained models, logs, and result visualizations such as accuracy and loss graphs. SSD storage is preferred for faster read and write operations.

4.3 Sentiment Analysis Using Machine Learning

The proposed system is developed using Python, TensorFlow, and Keras to perform sentiment analysis on textual data. The main objective of the system is to identify whether a given text expresses a positive or negative sentiment.

The workflow of the system includes the following steps:

1. Dataset Loading

The Amazon product reviews dataset is loaded and relevant columns such as review text and ratings are selected.

2. Data Preprocessing

Text data is cleaned by removing null values and neutral reviews. Reviews are labeled as positive or negative based on their rating score.

3. Text Tokenization and Padding

The textual data is converted into numerical form using a tokenizer. Sequences are padded to ensure uniform input length for the model.

4. Model Training

An LSTM-based deep learning model is trained using embedding layers, dropout for regularization, and a sigmoid activation function for binary classification.

5. Prediction and Evaluation

The trained model predicts sentiment for new unseen reviews. Performance is evaluated using accuracy score, confusion matrix, and classification report.

6. Result Visualization

Graphs for training accuracy, validation accuracy, training loss, and validation loss are generated to analyze model performance.

System Outcome :

The Sentiment Analysis system successfully converts raw textual reviews into meaningful sentiment insights. It helps organizations and users understand customer opinions, improve products, and make data-driven decisions efficiently.

4.4 Graphical Representation

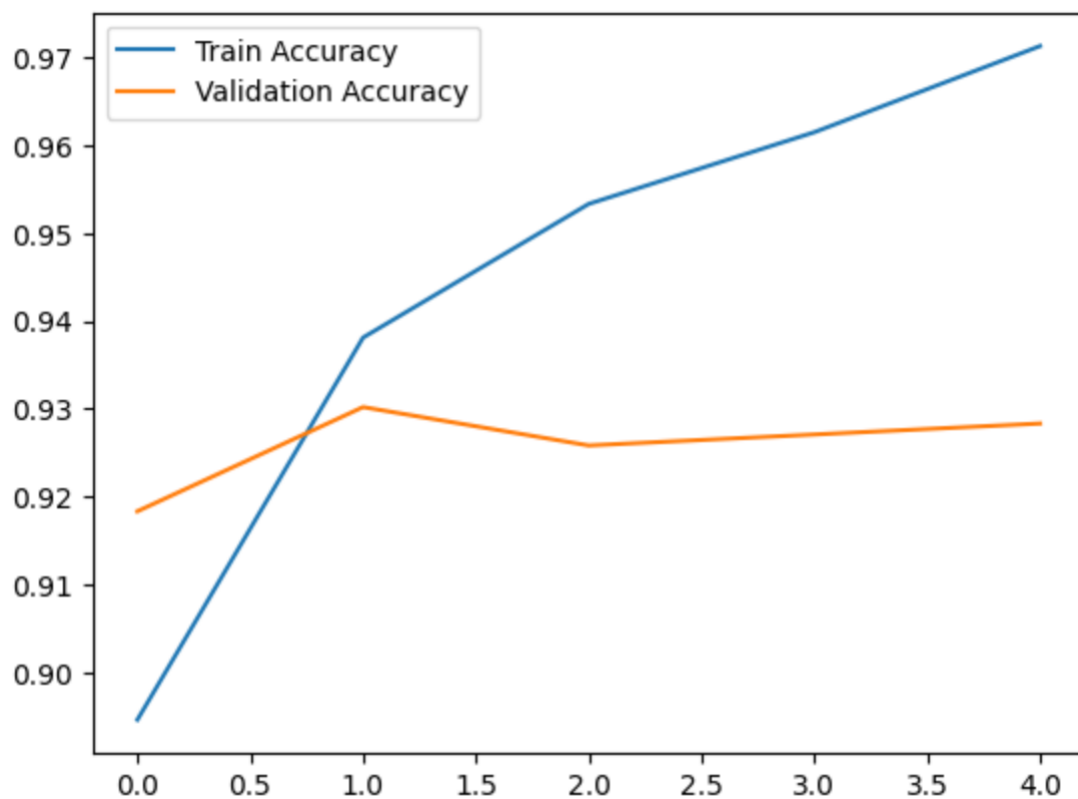


Fig 4.1 Train Accuracy vs Validation Accuracy

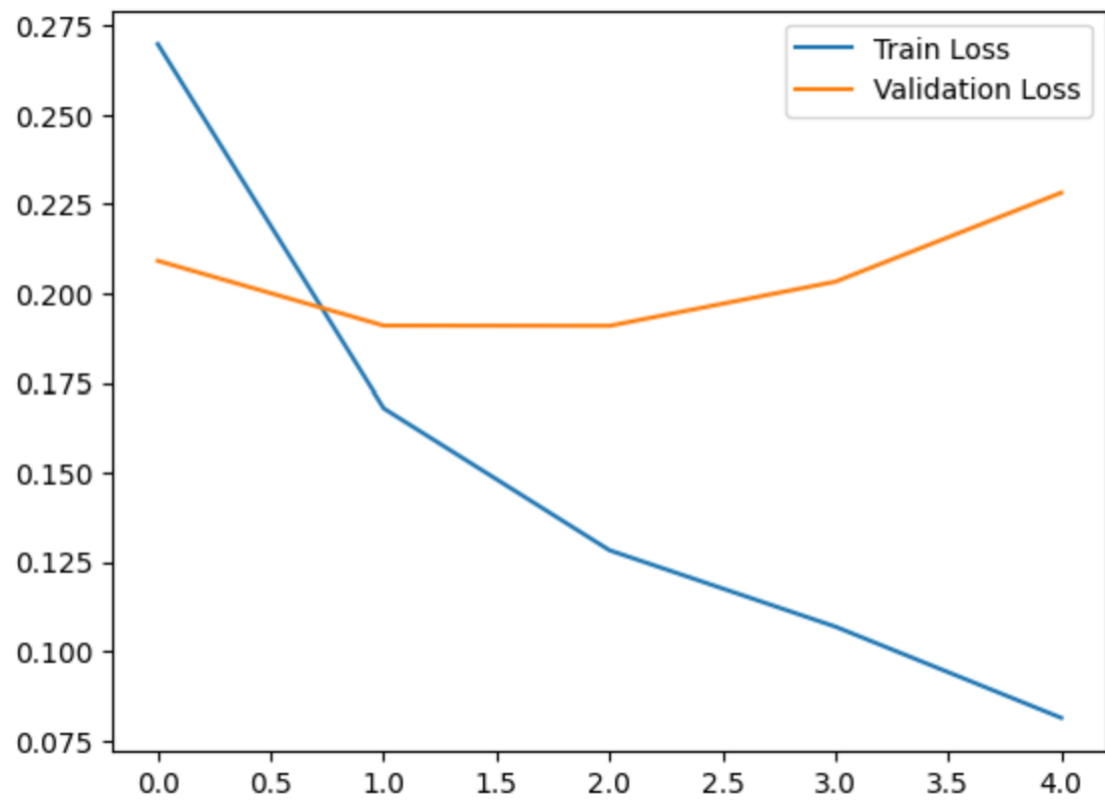


Fig 4.2 Train Loss vs Validation Loss

CHAPTER 5

5.1 Advantages of Sentiment Analysis Software

1. Automated Opinion Analysis

The system automatically analyses large volumes of customer reviews without human intervention. This saves time and effort compared to manual review analysis.

2. Efficient Handling of Large Datasets

The model efficiently processes thousands of Amazon product reviews, making it suitable for real-world applications where data size is large.

3. High Accuracy Using LSTM Model

The use of Long Short-Term Memory (LSTM) networks helps in capturing contextual relationships between words, leading to improved sentiment classification accuracy.

4. Real-Time Sentiment Prediction

The trained model can instantly predict sentiment for new user input reviews, making it useful for live feedback analysis.

5. Better Decision Making

Businesses can understand customer satisfaction levels and identify product strengths and weaknesses based on sentiment trends.

6. Scalable and Extendable

The system can be easily extended to include neutral sentiment, multi-class classification, or reviews from other domains like social media and blogs.

7. Visual Performance Evaluation

Graphs such as accuracy, loss curves, and confusion matrix help in clearly understanding model performance and reliability.

5.2 Disadvantages of Sentiment Analysis Software

1. Binary Classification Limitation

The current system classifies reviews only as positive or negative and does not handle neutral or mixed sentiments.

2. Dependency on Quality of Dataset

Model performance heavily depends on the quality and balance of the dataset. Biased or noisy data can reduce accuracy.

3. Difficulty in Understanding Sarcasm and Context

Sarcastic reviews or complex language structures may lead to incorrect sentiment prediction.

4. High Computational Requirement

Training deep learning models like LSTM requires significant computational resources, especially for large datasets.

5. Language Dependency

The system is trained only on English text and may not perform well for reviews written in other languages.

6. Training Time

Model training can be time-consuming, particularly when running on systems without GPU support.

5.3 Conclusion

This project successfully implements a Sentiment Analysis system using Natural Language Processing and an LSTM-based deep learning model to classify Amazon product reviews into positive and negative sentiments. The system effectively processes raw textual data through preprocessing, tokenization, and sequence padding to prepare it for model training. The use of LSTM enables the model to capture contextual and sequential information in text, resulting in accurate and reliable sentiment predictions.

The performance of the system is validated using accuracy metrics, confusion matrix, and graphical analysis of training and validation results. Additionally, the developed prediction function allows real-time sentiment analysis for new user inputs, making the system practical for real-world applications such as customer feedback evaluation and product analysis.

Although the system has limitations in handling sarcasm, neutral sentiments, and multilingual data, it provides a strong foundation for advanced sentiment analysis solutions. Overall, the project demonstrates the effective use of deep learning techniques to extract meaningful insights from large-scale textual data and supports informed decision-making in modern digital platforms.

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